Inverse Relations and Functions

Name _____

Algebra 2 Lesson 5-6 (fast track) Date _____ Period _____

Learning goal: I can find an equation of the inverse of a function, state the domain of the inverse, and use composition to verify that two functions are inverses.

Language goal: I can explain how a function and its inverse are related using the words *inverse*, *reflection*, *domain*, *range*, and *composition*.

Spanish: inverse function = *función inversa*; inverse relation = *relación inversa*; relation = *relación*.

On the quiz: Example 4 gets you (4) the inverse of a cube-root function. Example 5 gets you (5) which function is the inverse of $f(x) = 2x + 4$ and (2, bonus) sketching a line and its inverse (the graph in Example 5A). Full Savvas quiz only: (1) the table of an inverse and (3) restricting the domain of a parabola.

On the Topic 5 test: Question 12 asks for the inverse of a square-root function (Example 4). Question 14 asks for $f^{-1}(f(10))$ from a table (Example 5). Question 17 asks which facts about the inverse of $1 + \sqrt{x}$ are true (Examples 4 and 5). Questions 6 (a cubic's inverse), 9 (two tables) and 15 (a pipe's diameter) come from Examples 2, 1 and 6, which this packet skips.

The idea: the inverse exchanges the roles of the domain and range.

Tables. Switch the columns: $(2, 6)$ on f is $(6, 2)$ on f^{-1}.

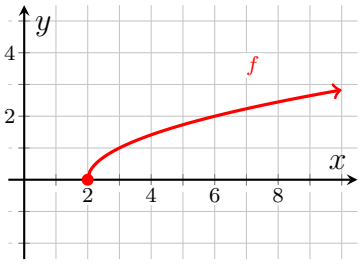
Algebra. Switch x and y, then solve for y: $y = x+9$, $x = y+9$, $y = x-9$.

Graphs. Reflect the graph across the line $y = x$.

Composition verifies inverses: $f(f^{-1}(x)) = x$ and $f^{-1}(f(x)) = x$.

Example 4: Find an Equation of an Inverse Function

A. Find an equation of the inverse function of $f(x) = \sqrt{x - 2}$.



No horizontal line crosses the graph more than once. Reflected over $y = x$, no vertical line will cross it more than once, so the inverse relation is a **function**.

	domain	range
f	$x \geq 2$	$y \geq 0$
f^{-1}	$x \geq 0$	$y \geq 2$

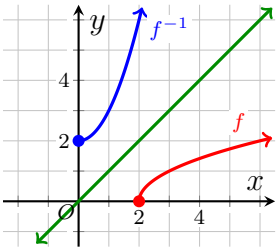
The **domain** and **range** trade places.

Switch the roles of x and y and solve for y.

1. $y = \sqrt{x - 2}$
2. $x = \sqrt{y - 2}$
3. $x^2 = y - 2$
4. $x^2 + 2 = y$

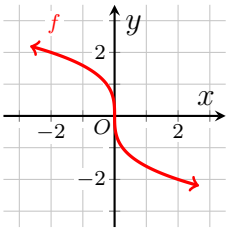
So $f^{-1}(x) = x^2 + 2$, $x \geq 0$.

Use f^{-1} notation only when the inverse is a function; the domain $x \geq 0$ is the range of f.



The graphs of f and f^{-1} are both functions and are **reflections** over the line $y = x$: $(3, 1)$ on f, $(1, 3)$ on f^{-1}.

B. Find an equation of the inverse function of $f(x) = -\sqrt[3]{4x}$.



No horizontal line crosses the graph more than once, so the inverse relation will be a function.

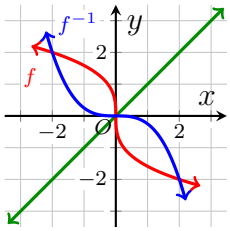
1. $y = -\sqrt[3]{4x}$

2. $x = -\sqrt[3]{4y}$

3. $x^3 = -4y$

4. $\dfrac{-x^3}{4} = y$

$f^{-1}(x) = \dfrac{-x^3}{4}$ for all real x.



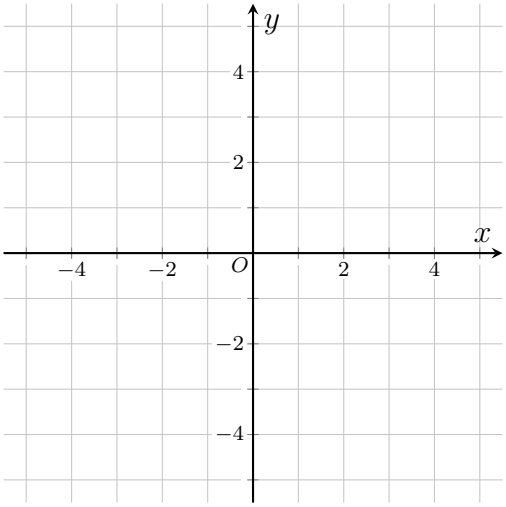
Say it: The inverse of $f(x) = \sqrt{x-2}$ is $f^{-1}(x) = x^2 + 2$ with $x \geq 0$, because the _____ of f becomes the _____ of the inverse. The graph of f^{-1} is the _____ of the graph of f across the line _____.

Try It 4: Write the inverse of a cube-root function

Let $f(x) = 2 - \sqrt[3]{x+1}$.

a. Sketch the graph of f.
Hint: the graph bends at $(-1, 2)$; find $f(-2)$ and $f(0)$ too.

x	-2	-1	0
$f(x)$			



b. Verify that the inverse will be a function and write an equation for $f^{-1}(x)$.

_____

$y = 2 - \sqrt[3]{x+1} \quad \rightarrow \quad x =$ _____

_____

_____

_____

$f^{-1}(x) =$ _____

Say it: No horizontal line crosses the graph of f more than once, so the _____ of f is a function. Its _____ is the range of f, and its graph is the _____ of the graph of f across $y = x$.

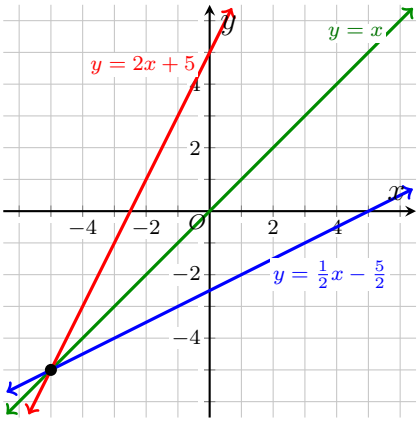
Example 5: Use Composition to Verify Inverse Functions

A. What is the inverse of $f(x) = 2x + 5$, and how can you verify this?

Switch x and y and solve for y:
$y = 2x + 5 \;\rightarrow\; x = 2y + 5,\;\; 2y = x - 5,\;\; y = \frac{1}{2}x - \frac{5}{2}$.
So the inverse of f is $g(x) = \frac{1}{2}x - \frac{5}{2}$. To be inverse functions, $(f \circ g)(x) = x$ and $(g \circ f)(x) = x$.

$$\begin{aligned}(f \circ g)(x) &= f(g(x)) \\ &= 2\big(g(x)\big) + 5 \\ &= 2\left(\tfrac{1}{2}x - \tfrac{5}{2}\right) + 5 \\ &= x - 5 + 5 \\ &= x\end{aligned} \qquad \begin{aligned}(g \circ f)(x) &= g(f(x)) \\ &= \tfrac{1}{2}\big(f(x)\big) - \tfrac{5}{2} \\ &= \tfrac{1}{2}(2x + 5) - \tfrac{5}{2} \\ &= x + \tfrac{5}{2} - \tfrac{5}{2} \\ &= x\end{aligned}$$

Both compositions are x, so $f(x) = 2x + 5$ and $g(x) = \frac{1}{2}x - \frac{5}{2}$ are inverses. You can also verify it by graphing: the graph of g is the **reflection** of the graph of f across $y = x$.



B. Are the two functions $f(x) = x^2 + 5$ and $g(x) = \sqrt{x} - 5$ inverses of each other?

$$\begin{aligned}(f \circ g)(x) &= f(g(x)) \\ &= \big(g(x)\big)^2 + 5 = (\sqrt{x} - 5)^2 + 5 \\ &= x - 10\sqrt{x} + 25 + 5 = x - 10\sqrt{x} + 30\end{aligned} \qquad \begin{aligned}(g \circ f)(x) &= g(f(x)) \\ &= \sqrt{f(x)} - 5 \\ &= \sqrt{x^2 + 5} - 5\end{aligned}$$

Neither composition simplifies to the identity function x, so the functions are **not** inverses.

Say it: In part A, f and g are inverses because the _____ $f(g(x))$ and $g(f(x))$ both simplify to _____. In part B, neither one does, so f and g are not _____.

Try It 5: Use composition to check two functions

Use composition to determine whether f and g are inverse functions.

a. $f(x) = \frac{1}{4}x + 7,\;\; g(x) = 4x - 7$

$(f \circ g)(x) = f(4x - 7) =$ _____

_____

_____

_____

Inverses? _____

b. $f(x) = \sqrt[3]{x - 1},\;\; g(x) = x^3 + 1$

$(f \circ g)(x) =$ _____

_____

$(g \circ f)(x) =$ _____

_____

Inverses? _____

Say it: In part ____, both _____ simplify to x, so f and g are _____.

IXL: Practice for the quiz

Go to IXL, Algebra 2. Type the code in the search box. Do the starred skills first.

Code	Skill	Feeds
ZRQ ★	Find inverse functions and relations	Quiz 4: switch and solve a cube-root function (Example 4); also Quiz 2 (bonus)
9KT ★	Identify inverse functions	Quiz 5: is g the inverse of f? (Example 5, composition)
K2W ★	Graphs of inverse functions	Quiz 2 (bonus): reflect a line across $y = x$ (Example 5A's graph)
YLX	Find values of inverse functions from tables	Optional: Topic test Question 14; Quiz 1 (full Savvas quiz only)
UAJ	Graph square root functions	Warm-up for Example 4A: graph a square-root function

SmartScore 70 or above on each starred (★) skill replaces your quiz score.

Exit question: Undo a 4th power (DOK 3)

Higher Order Thinking What is the inverse operation of raising a number to the 4th power? How can you use the inverse operation of a number raised to the 4th power to find the inverse of the function $f(x) = x^4 - 1$? Is the inverse of f a function? Explain.

Before the quiz, finish these sentences.

To find an inverse, I switch x and y and solve for y. The _____ of f becomes the _____ of f^{-1}.

The graph of the inverse is the _____ of the graph of f across the line _____.

To check that f and g are inverses, I use _____: both $f(g(x))$ and $g(f(x))$ must equal _____.